

# LECTURE: ##17

Date

## HOW TO DESIGN HARDWARE PRODUCTS (Hosain Rahman)

### BEAUTY AND ENGINEERING IN SERVICE OF A BETTER LIFE

- \* The world around us have been a little bit of a mess.
- \* There is an desperate need around all of organizing principles around it.
- \* Think about where is the world going?
  - ↳ Build out the things that the users are going to look around desperately.

\*

### THINKING ABOUT INDIVIDUAL USERS

- \* Context Engine: You can tell the smart thermostat in your nest can tell that if you are cold or hot.
- \* You need to be great at everything that will primarily include the full stack itself.

\* \* Intersection for the first time for building the company and working on those variables.

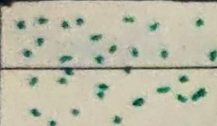
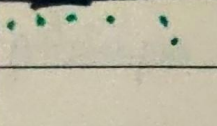
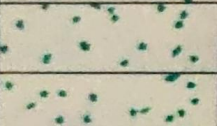
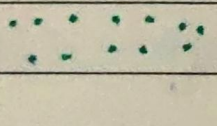
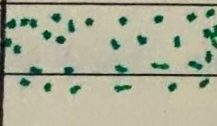
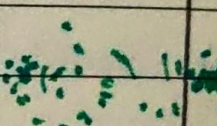
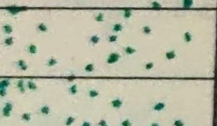
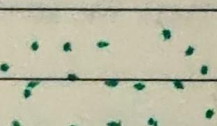
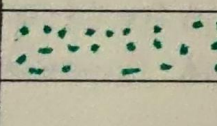
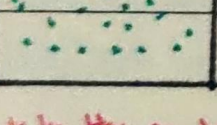
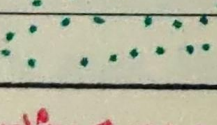
HARDWARE,  
SOFTWARE & SERVICES  
AND DATA

- \* You have to take time because the iteration cycles are much more greener.

# EVERYTHING IS A SYSTEM

\* Think about it across the whole sector.

## HOW WE CREATE

| Exploration                                                                       | Early validation                                                                   | CONCEPT                                                                           | PLANNING                                                                           | DEV. | CONV INNOV. |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------|-------------|
|   |   |                                                                                   |                                                                                    |      |             |
|   |   |  |  |      |             |
|   |   |  |                                                                                    |      |             |
|  |  |                                                                                   |                                                                                    |      |             |

\* In the explorative stage you must be imaginative, really broad and then understand that where does the brand stand for.

## Unbridled Imagination

\* Dreaming

\* Futuristic

\* Disruptive innovation

\* Science project-y

\* Rigorously defined building blocks

\* Building from inspiration and insights

## EARLY VALIDATION

\* Proving your thesis like a PhD

\* Exploring the system.

\* Outlining a story.

## CONCEPT

Bg human-centric concepts.

\* Starting on a journey.

- \* Excitement about what's possible
- \* Pairingly experiences to brand stories.
- \* Asking big questions.
- \* Selling the experience.

## PLANNING

- \* Emergence of a product.
- \* No turning back
- \* Building belief
- \* A bit coloring.
- \* Made-off conversations and creative problem solving.
- \* Breakthroughs.
- \* Lots of diverse input.

## DEVELOPMENT:

- \* Bringing it to life.
- \* Anchoring the relay
- \* Having a shared understanding of what, why and how.
- \* Laser-focused problem solving.
- \* Pride in craftsmanship.



## CONTINUED INNOVATION

- \* Feeding the fire.
- \* Going from imagination to execution on specific challenges.
- \* Creating a challenge within the world.
- \* Developing strategies to achieve focused goals.
- \* Leveraging what we have created to do even more.

# EXPLORATION

Date

1. What you are doing?
2. What drives it?
3. The tools we use.
4. Who is involved?
5. What is required from E staff?



LOOK OUT FOR INVESTMENTS



FITS INTO OUR STRATEGIC VISION; IS FEASIBLE TO MAKE?  
BUSINESS VIABILITY; EARLY EXPERIENCE BRIEFS

## \* CONCEPT

- \* Everyone thinks about the conventional designs and the general creators.
- \* the people try to really drive the vision and the experiences with the tools that are going to be responsible.
- \* where the world is moving and how is that going to be an evolution of the functions.



PLANNING

PRIORITIZED FEATURES, FUNCTIONALITY, AND MINIMUM Viable;  
BUSINESS PLAN AND PRODUCT ROADMAP.



Date

## DEVELOPMENT

DESIGN SIGN-OFF; ENGINEERING HARDWARE & SOFTWARE SIGN-OFF;

HOSEAN SIGN-OFF.

## THE FRAMEWORK

CREATING RICH, CONTINUOUS & SIGNATURE EXPERIENCES

| <u>WHYS</u>                                                      | <u>THEMES</u>                                                                 | <u>PODS</u>                                               | <u>HERO/SIDEKICK</u>                                                                           |
|------------------------------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------------------|
| A clear and memorable articulation of human & business problems. | Experience-based story tracks that translate 'WHYS' into actionable concepts. | Cross func teams empowered to experiment and own a theme. | Scenarios that elevate concepts to signature solutions: ownable, system oriented, intentional. |
| <u>FEATURES</u>                                                  |                                                                               |                                                           |                                                                                                |
| The building blocks of core elements of an experience: scalable. |                                                                               |                                                           |                                                                                                |

WHAT IS THE USER PROBLEM THAT ONCE WE SOLVE, USERS CAN'T LIVE WITHOUT

\* EX. WORKS ARE EXTREMELY INNOVATIVE

\* You are going to selling things within an interesting strategy.

\* The Experience Continuum is something that must be considered.

## EXPERIENCE CONTINUUM

0----- TODAY

0----- TOMORROW

0----- FUTURE

\* We create experiences, not just hardware.

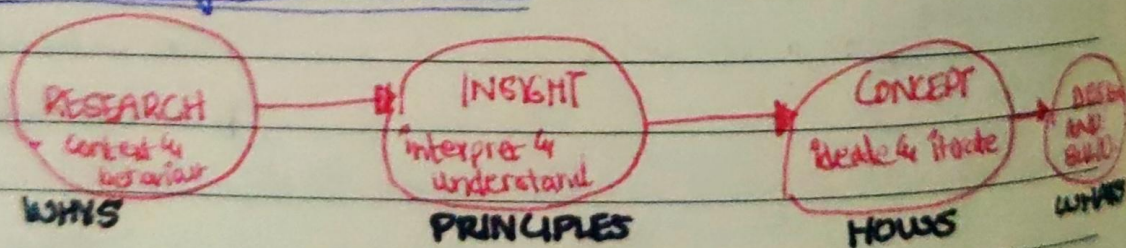
\* Where you need to innovate?

\* The focus must be on the things that are going to be magical to the users.

\*

ALL OF THESE THINGS BECOME A  
BRIEF FOR YOUR ENGINEERING TEAM  
OR YOUR DESIGN TEAM.

## Understanding the User process



\* You have to separate that what are the questions you are going to ask that are going to help make you smarter about your thesis vs trying to get something to validate.

\* UP24

⇒ work on the opposites and how you are going to be working on those creations.

\* The behaviour loop.

\* working on the hero principle.

\* Your last chance rush.

\* Constraints help you to push you to the right choices.

Q. When you have a number of features and functions that you're trying to build, how do you look it at a system level?

\* You just don't think about in one silo. You have to force a lot when it's a small team because you are sitting around the table.

\* Everyone has to share what their pains are.

\* Think about the ways they need to accomplish itself for the danger points, liabilities and the things.

\* A system is just a mindset

Inter-dependence  
Collaboration

\* Within everything there's all the system.

\* we can make the world everything around you smarter

↳ Different categories of the dynamics and the problem solving skills are going to be different.

↳ set of underlying the problems we are going to solve

↳ go straight to planning and development and understanding the spirit innovation.

\* Communication to understand those leadership roles.